Data Glacier

Your Deep Learning Partner

Project: Bank Marketing Campaign

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Github link:
https://github.com/priyanjalipatel/Data_Glacier_Final_Project

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Background

- **Business Problem**

❖ ABC Bank wants to increase **term deposit subscriptions**.

❖ Marketing campaigns (phone calls) are **costly and time-consuming**.

❖ Goal → **Predict customers most likely to subscribe** so bank can:

  ➢ Save resources 💰

  ➢ Improve targeting 🎯

Objective:

ABC Bank wants to sell it's term deposit product to customers and before launching the product they want to develop a model which help them in understanding whether a particular customer will buy their product or not (based on customer's past interaction with bank or other Financial Institution).

Data Exploration

- Source: UCI Bank Marketing Dataset (~45,000 records).
- Target Variable: y (yes/no for subscription).
- Features:
  - Demographics (age, job, marital, education, etc.)
  - Contact info (communication type, campaign outcomes)
  - Historical data (pdays, previous, etc.).

Challenges

- Imbalanced dataset (~90% No, ~10% Yes).
- Some misleading features (duration not practical in real-time).
- Need interpretable + scalable models.

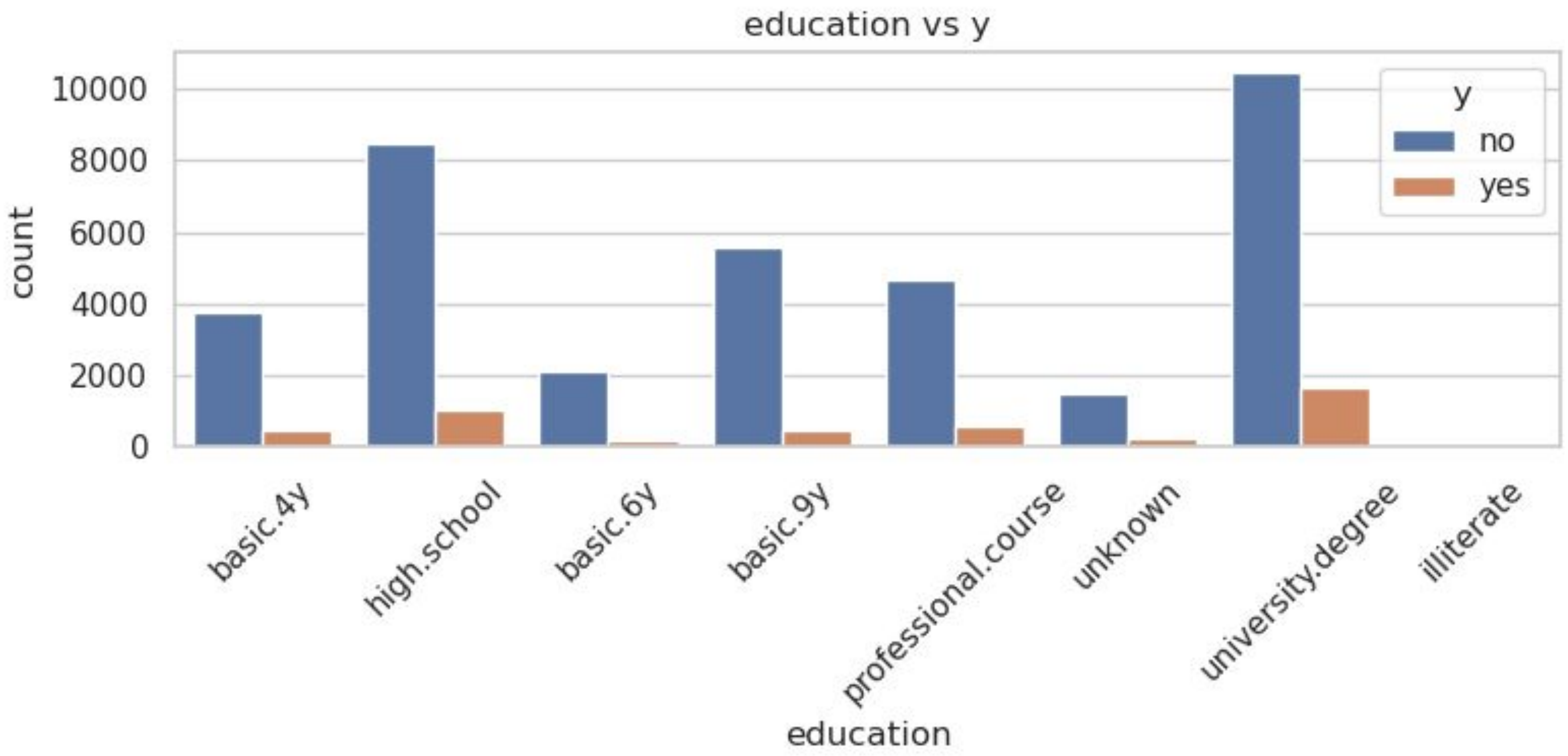
Approach

1. Business Understanding
2. Data Exploration (EDA)
3. Preprocessing (scaling, encoding, handling imbalance)
4. Model Building (Logistic Regression, Tree-based, Ensemble)
5. Evaluation (ROC-AUC, F1, Recall, Confusion Matrix)
6. Recommendations for Deployment

2. EDA Highlights

- Most customers contacted multiple times → low success rate.
- pdays = 999 → indicates customer not contacted before (special case).
- Education & job type have clear influence on subscription.

2. Exploratory Data Analysis (EDA)



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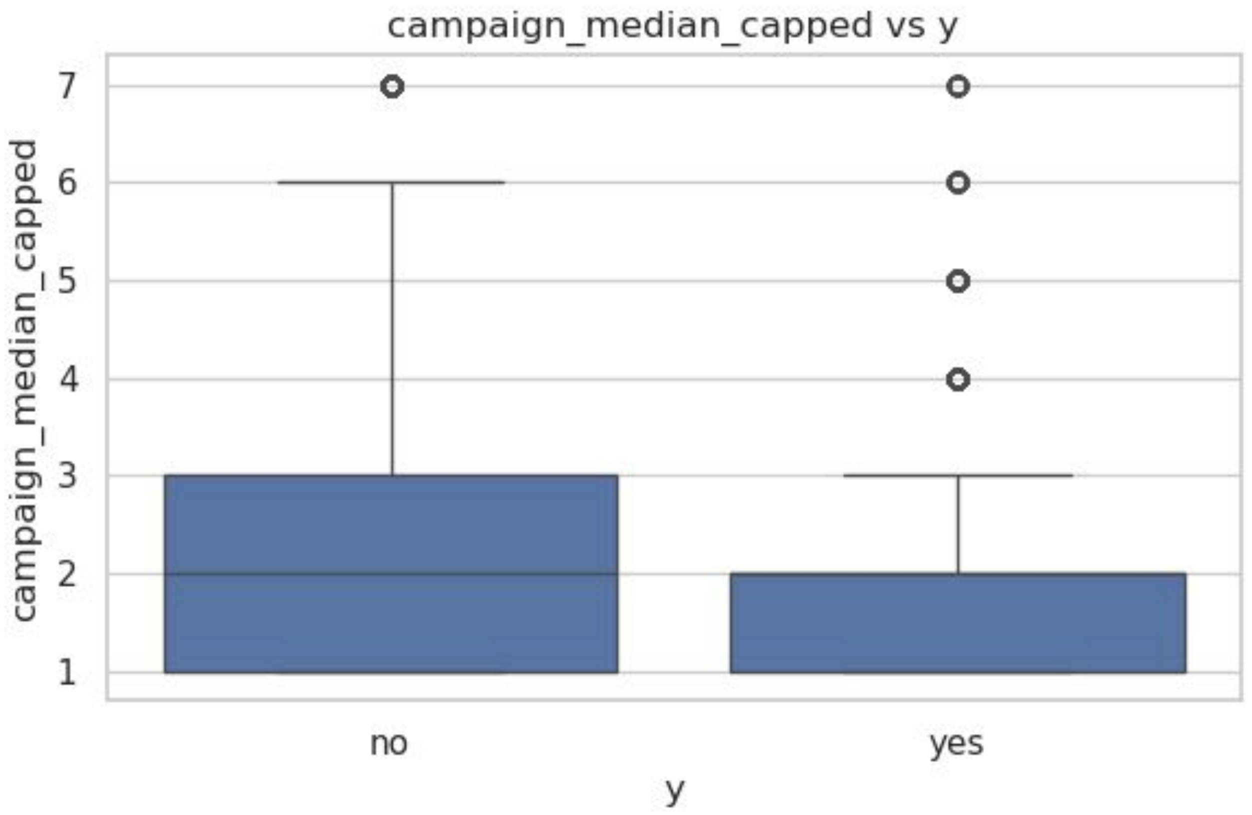

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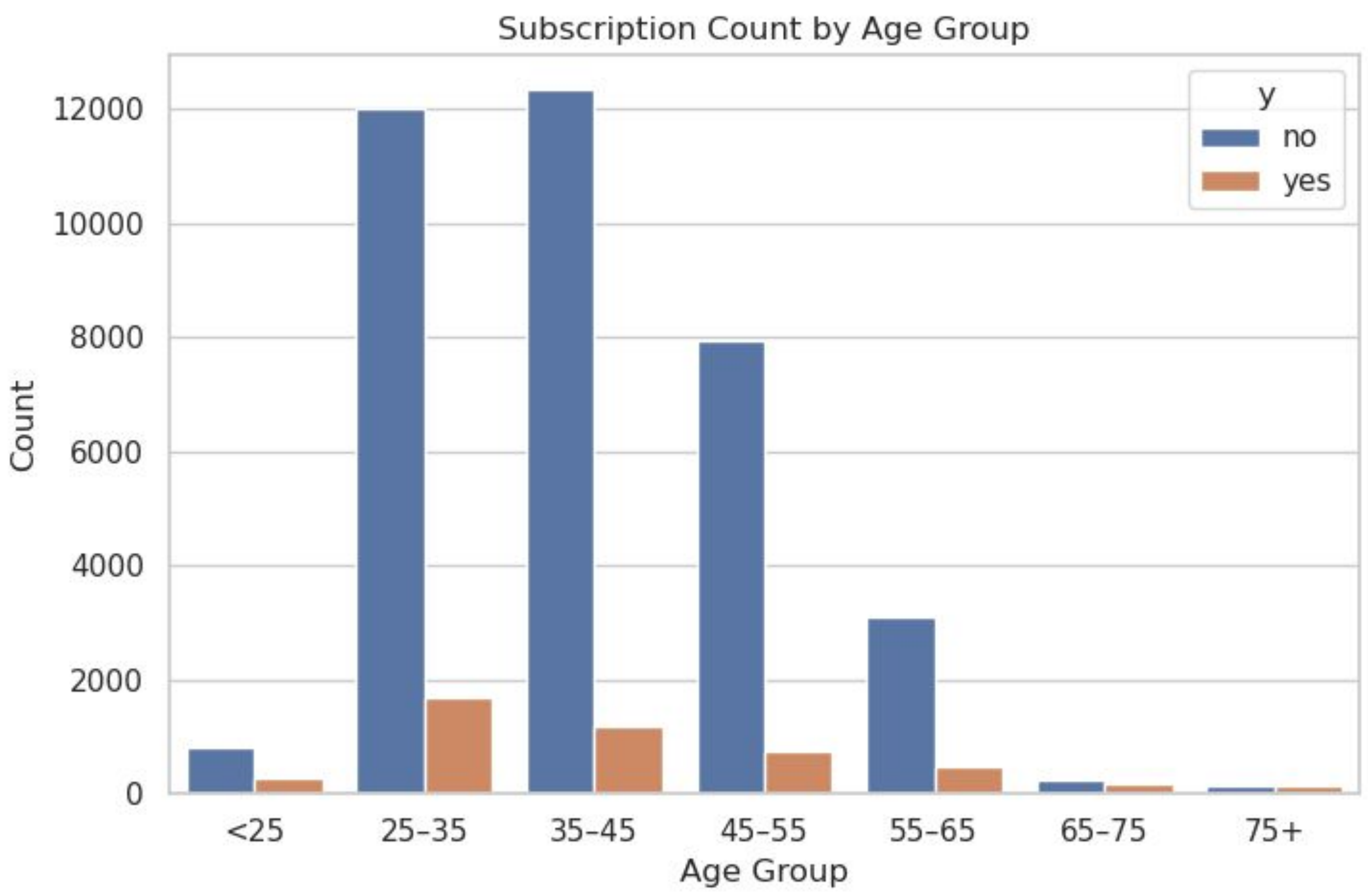

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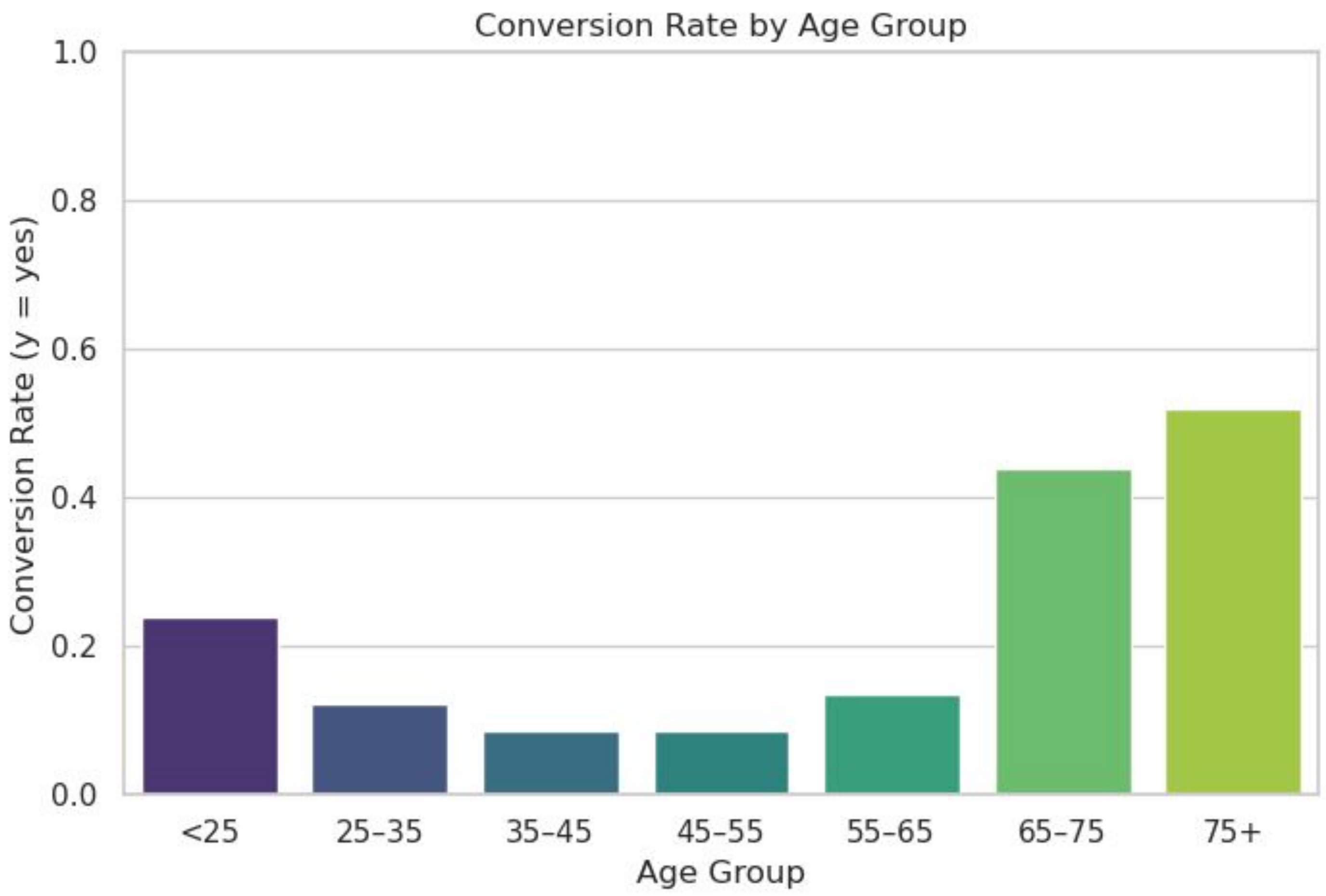
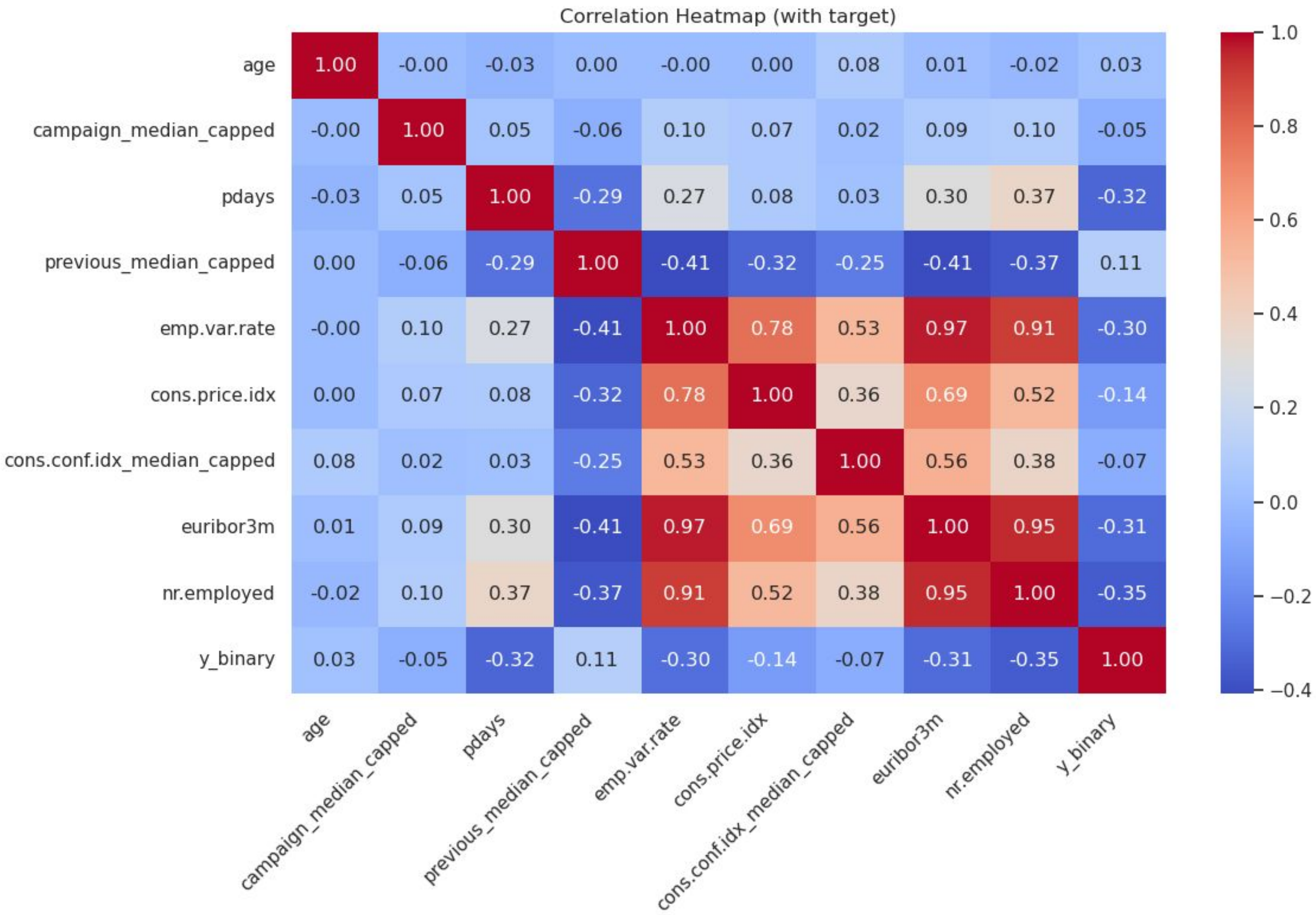


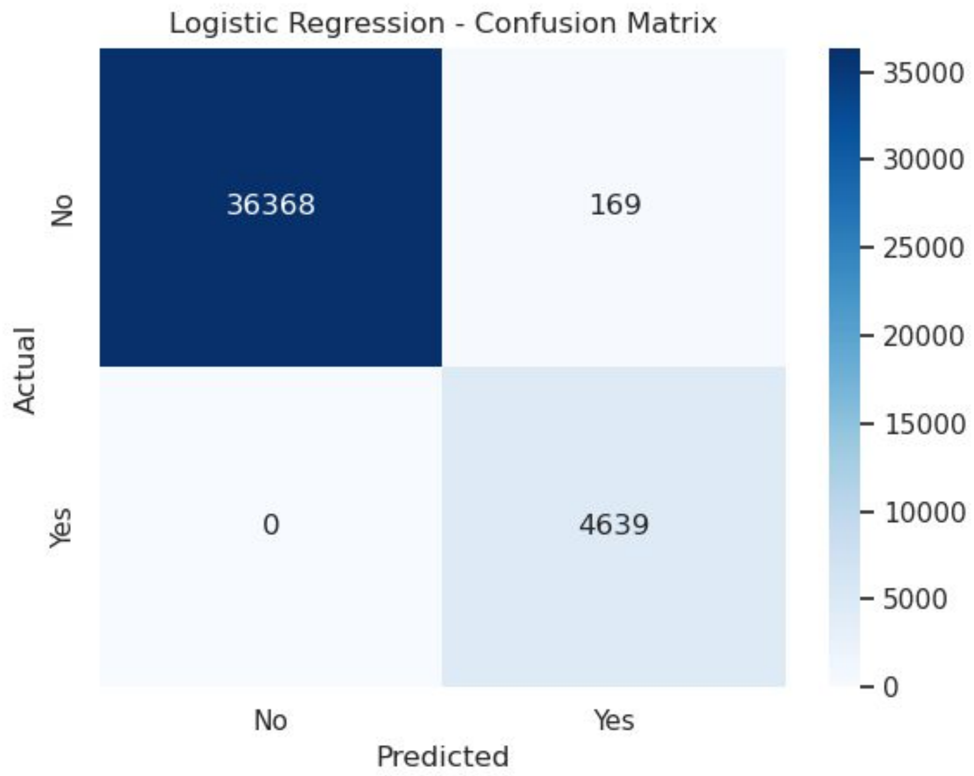
Correlation heatmap

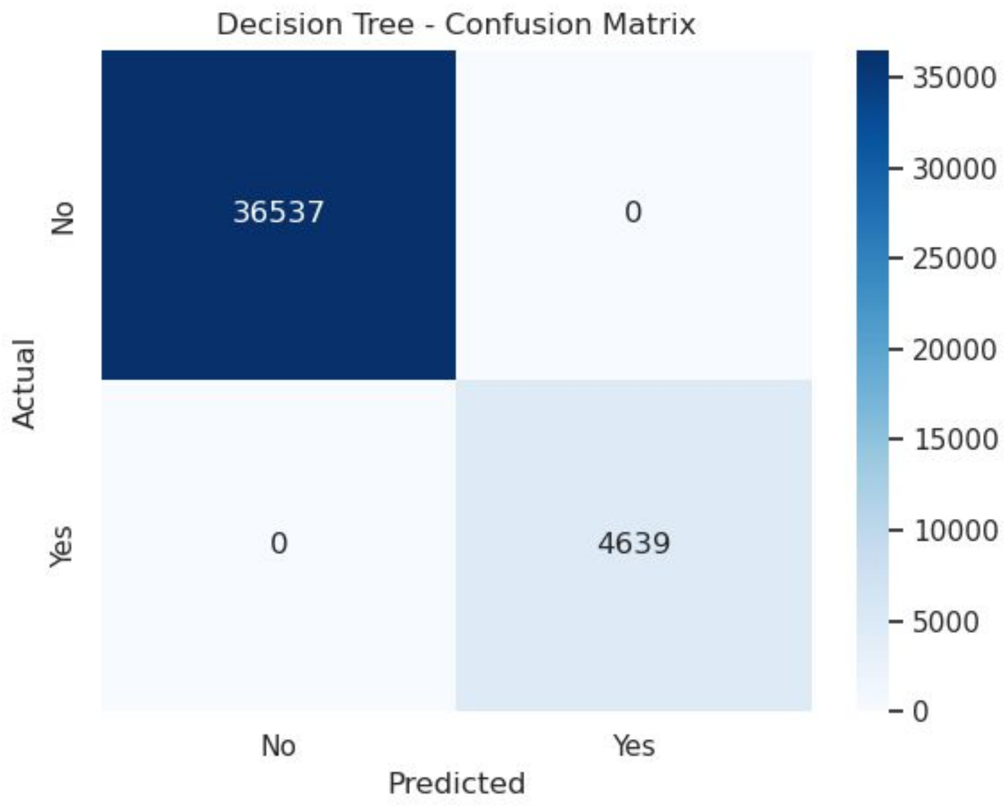


Models Compared

- Logistic Regression
- Decision Tree
- Naive Bayes
- Random Forest
- Gradient Boosting
- Evaluation Metrics: Accuracy, ROC-AUC, Recall, Precision

Models Compared





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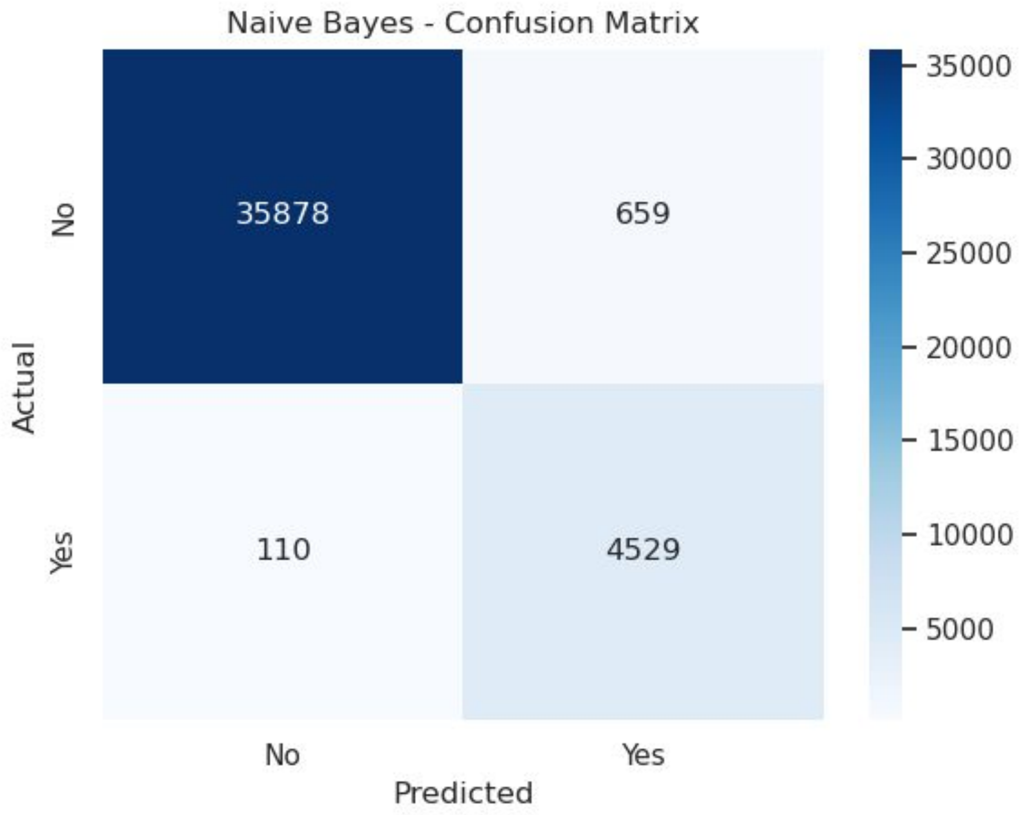

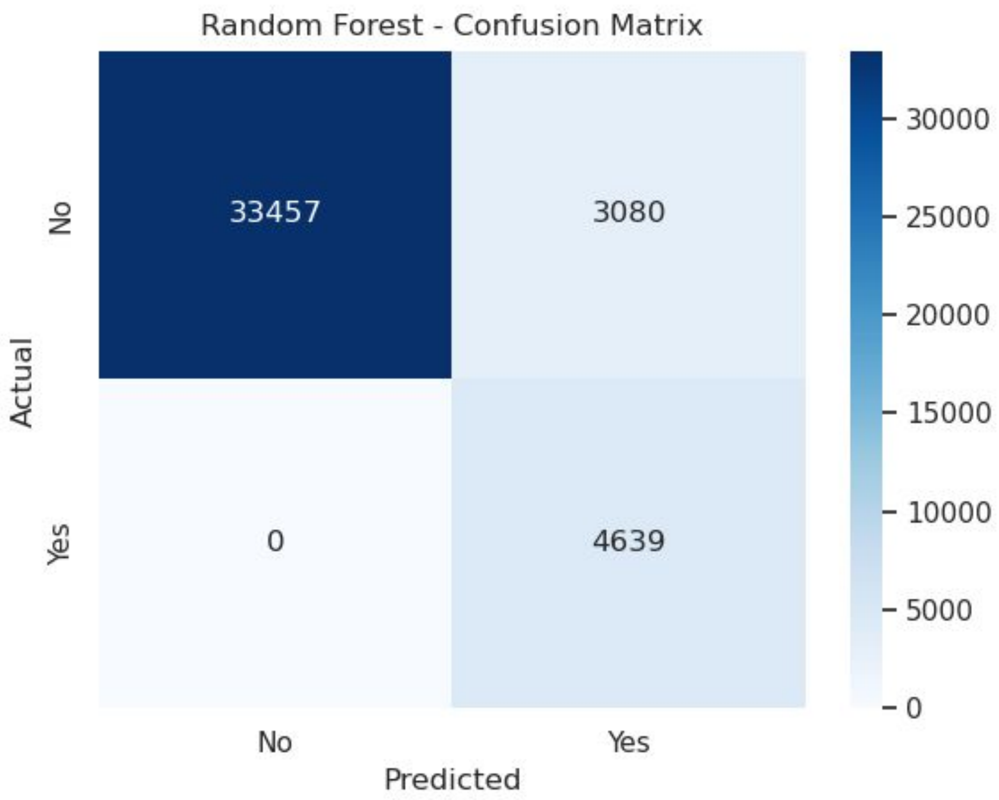

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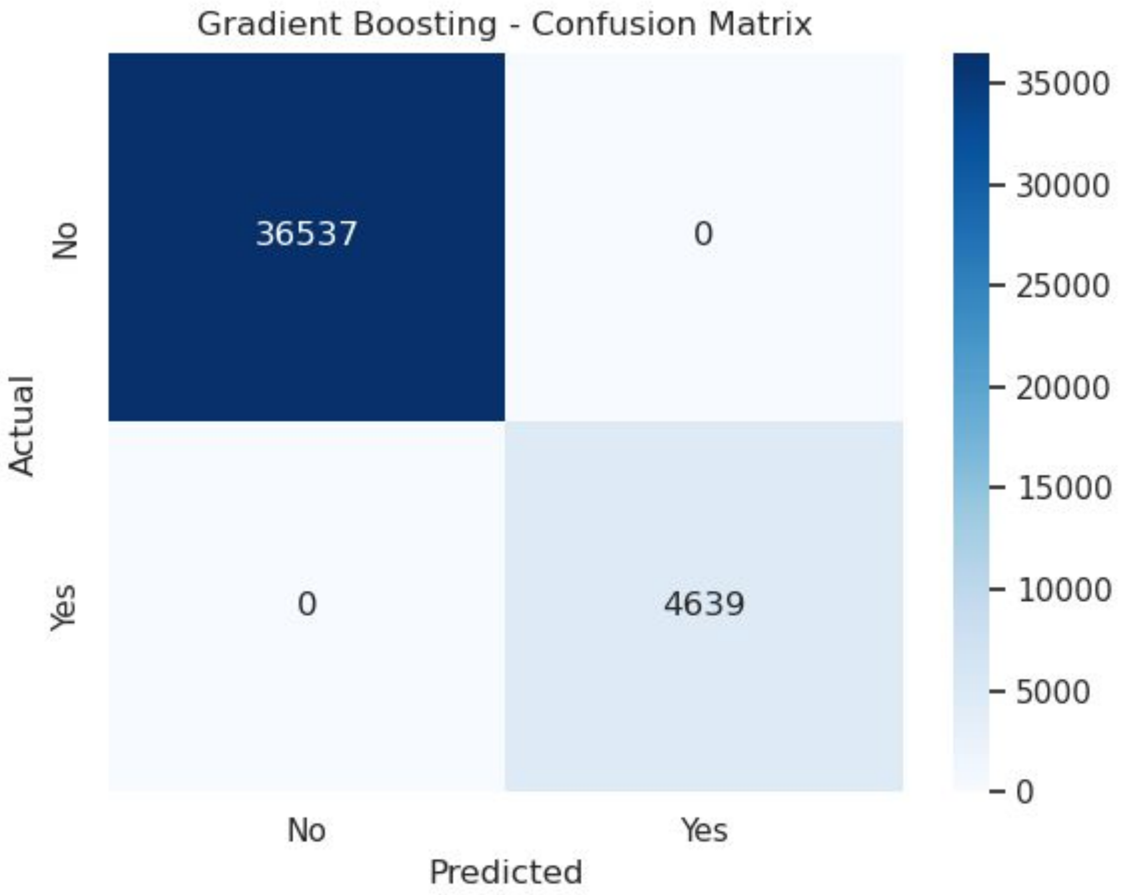

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Models Compared

Logistic Regression

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Accuracy: 0.9959

ROC-AUC : 1.0000
F1-Score: 0.9846

Decision Tree

==============

Accuracy: 1.0000

ROC-AUC : 1.0000

F1-Score: 1.0000

Naive Bayes

============

Accuracy: 0.9813

ROC-AUC : 1.0000

F1-Score: 0.9418

Random Forest

==============

Accuracy: 0.8805

ROC-AUC : 0.9999

F1-Score: 0.8547

Gradient Boosting

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Accuracy: 1.0000

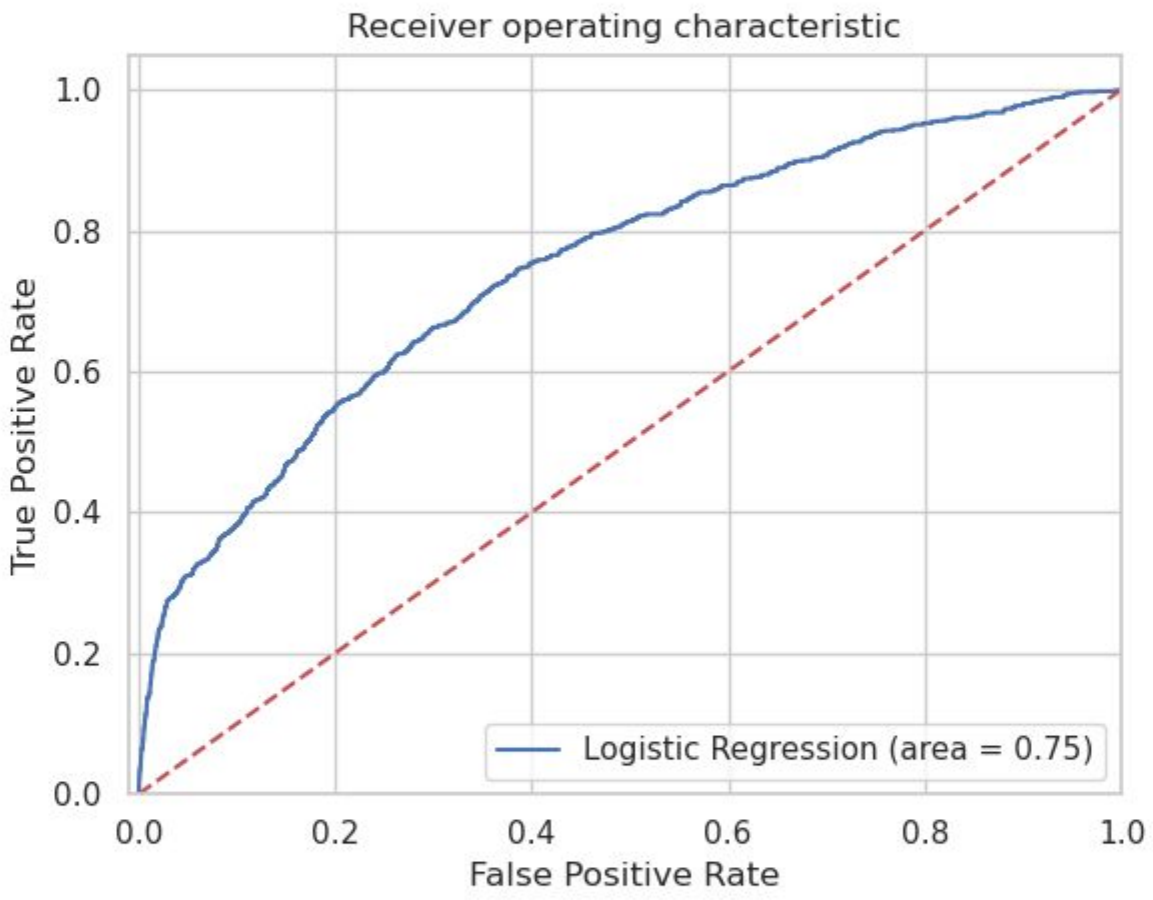
ROC-AUC : 1.0000

F1-Score: 1.0000

- Decision Tree and Gradient Boosting showed 100% accuracy, likely due to overfitting or data leakage (e.g., use of duration feature). These models were excluded from final selection to ensure fair evaluation.

LogisticRegression is the best model

ROC-AUC

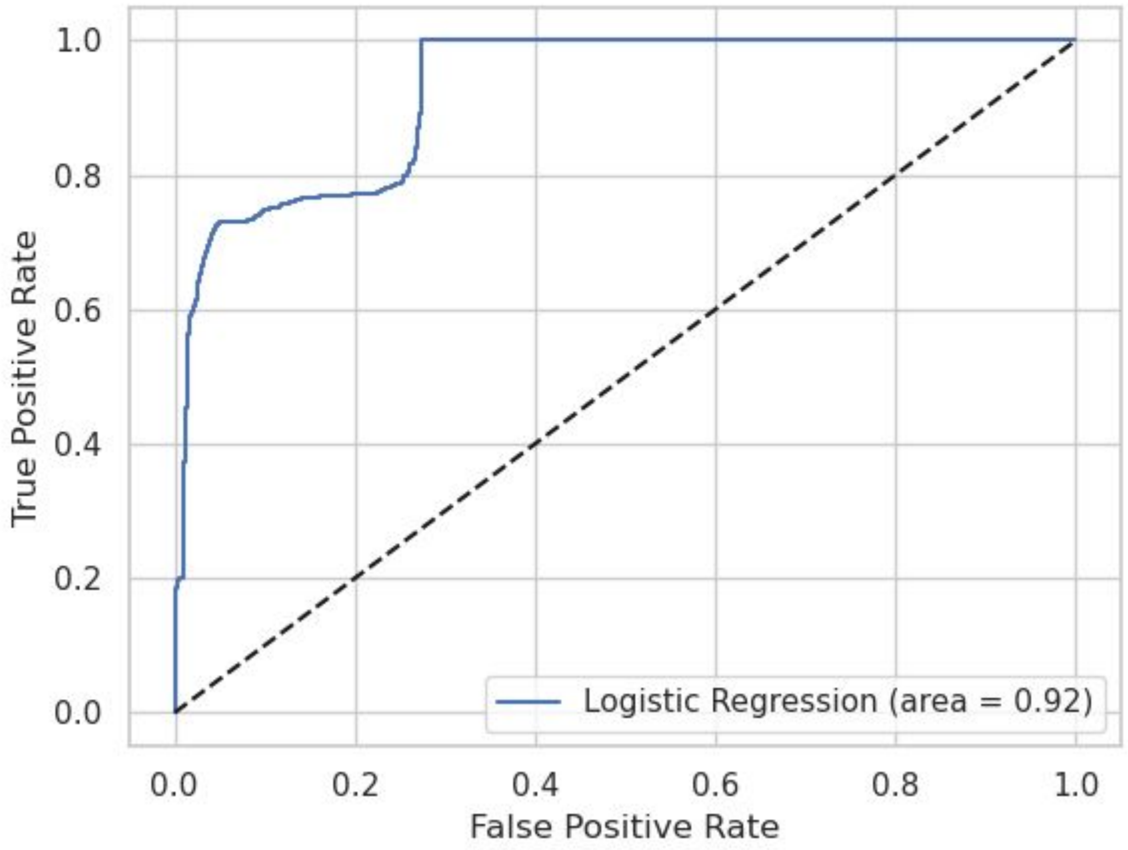


Imbalance Handling

Imbalance Handling

- Used **Class Weights** & **SMOTE**.
- Improved Recall for "Yes" class (subscribers).
- Tradeoff: Slight drop in overall accuracy but better for business.

ROC-AUC



1.0
0.8
0.6
0.4
0.2
0.0
True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate
Logistic Regression (area = 0.92)

Key Insights

- Logistic Regression: Best balance between interpretability & performance.
- Recall improvement crucial for campaign success.
- ROC-AUC ~0.92 (LogReg) → good discrimination ability.
- **Logistic Regression (Balanced)**
    a. Accuracy: **90%**
    b. Recall (Yes): **73%** (good improvement for minority class)
    c. Precision (Yes): **53%**
    d. ROC-AUC: **0.92**

ROC-AUC

ROC-AUC meaning

- **ROC curve** plots **True Positive Rate (Recall)** vs. **False Positive Rate** at different thresholds.
- **AUC (Area Under Curve)** is a single number summarizing the curve:
  - **1.0 → perfect model**
  - **0.5 → random guessing**
  - Higher is always better.

So if you see **ROC-AUC = 0.75 vs 0.92**:

- **0.75** → pretty good model (better than random, decent separability).
- **0.92** → only slightly better than random, weak predictive power.

So **0.92 is better**.

Business Impact

- More efficient marketing campaigns.
- Save costs by avoiding customers unlikely to subscribe.
- Targeted campaigns → higher conversion rate.

Conclusion & Recommendations

- Predict if a client will subscribe to a term deposit (Yes/No) from Portuguese bank marketing campaign data.

Best Model

- **Logistic Regression (Balanced)**

Insights

- Dataset was **imbalanced** (fewer "Yes" cases).
- Balanced Logistic Regression captured more subscribers compared to other models.
- Some false positives exist, but recall improvement ensures fewer missed subscribers.

Recommendation

- Use the balanced Logistic Regression model for deployment.
- Prioritize **recall** to identify more potential customers.
- Next steps: collect more data, explore boosting models (XGBoost, LightGBM), and monitor model in production.

Thank You

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